

Lab 1

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Help

Query History

1 BEGIN;
2 UPDATE film SET rental_rate = 4.99 WHERE film_id = 1;
3 -- Lúc này Session A đang giữ X-Lock trên dòng có film_id = 1
4

Data Output

Messages

Notifications

UPDATE 1
Query returned successfully in 83 msec.

Query

Query History

1 BEGIN;
2 UPDATE film SET rental_rate = 5.99 WHERE film_id = 1;
3 -- BỊ CHẪN NGAY LẬP TỨC!
4 -- Đang phải chờ Session A nhả khóa.

Data Output

Messages

Notifications

Data Output

Messages

Notifications

UPDATE 1
Query returned successfully in 83 msec.

Query

Query History

1 BEGIN;
2 UPDATE film SET rental_rate = 4.99 WHERE film_id = 1;
3 COMMIT;

Data Output

Messages

Notifications

Data Output

Messages

Notifications

WARNING: there is already a transaction in progress
COMMIT
Query returned successfully in 271 msec.

Query

Query History

1 BEGIN;
2 UPDATE film SET rental_rate = 5.99 WHERE film_id = 1;
3 COMMIT;

Data Output

Messages

Notifications

Data Output

Messages

Notifications

WARNING: there is already a transaction in progress
COMMIT
Query returned successfully in 136 msec.

Lab2:

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Help

Query History

1 BEGIN;
2 SELECT rental_rate FROM film WHERE film_id = 10;
3 -- Kết quả trả về: 4.99 (Shared Lock nhả ra ngay sau khi đọc xong)

Data Output

Messages

Notifications

Showing rows: 1 to 1
rental_rate numeric(4,2)
1 4.99

Query

Query History

1 BEGIN;
2 UPDATE film SET rental_rate = 0.99 WHERE film_id = 10;
3 COMMIT;
4 -- Sửa đổi và lưu dữ liệu trong khi Session A vẫn đang active.

Data Output

Messages

Notifications

Data Output

Messages

Notifications

COMMIT
Query returned successfully in 125 msec.

Query

Query History

1 BEGIN;
2 UPDATE film SET rental_rate = 0.99 WHERE film_id = 10;
3 COMMIT;
4 -- Sửa đổi và lưu dữ liệu trong khi Session A vẫn đang active.

Data Output

Messages

Notifications

Data Output

Messages

Notifications

COMMIT
Query returned successfully in 125 msec.

Lab3:

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Help

Query History

1 BEGIN;
2 UPDATE film SET rental_rate = 2.99 WHERE film_id = 100;

Data Output

Messages

Notifications

UPDATE 1
Query returned successfully in 144 msec.

Query

Query History

1 BEGIN;
2 UPDATE film SET rental_rate = 2.99 WHERE film_id = 200;

Data Output

Messages

Notifications

Data Output

Messages

Notifications

UPDATE 1
Query returned successfully in 94 msec.

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Query Query History

1 UPDATE film SET rental_rate = 3.99 WHERE film_id = 200;

2 -- Lệnh bị chặn (Đang chờ Session B nhả khóa)

Data Output Messages Notifications

UPDATE 1

Query returned successfully in 3 secs 425 msec.

Query Query History

1 UPDATE film SET rental_rate = 3.99 WHERE film_id = 100;

2 -- BÙM! LỖI: deadlock detected

Data Output Messages Notifications

ERROR: deadlock detected

Process 760 waits for ShareLock on transaction 1513; blocked by process 16628.

Process 16628 waits for ShareLock on transaction 1514; blocked by process 760.

SQL state: 40P01

Detail: Process 760 waits for ShareLock on transaction 1513; blocked by process 16628.

Process 16628 waits for ShareLock on transaction 1514; blocked by process 760.

Hint: See server log for query details.

Context: while locking tuple (5,12) in relation "film"

Lab4:

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Query Query History

1 BEGIN;

2 SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;

3 -- Chủ động khóa dòng này trước bằng FOR UPDATE

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

	inventory_id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2	2006-02-15 10:09:17

Query Query History

1 BEGIN;

2 SELECT * FROM inventory WHERE inventory_id = 10 FOR UPDATE;

3 -- BỊ CHẶN NGAY LẬP TỨC (do A đang giữ khóa dòng)

Data Output Messages Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

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Query Query History

1 INSERT INTO rental (rental_date, inventory_id, customer_id, staff_id)

2 VALUES (NOW(), 10, 1, 1);

3 COMMIT;

Data Output Messages Notifications

COMMIT

Query returned successfully in 233 msec.

Query Query History

1 ROLLBACK;

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 94 msec.